

What is USB (Universal Serial Bus)?

A Universal Serial Bus (*USB*) is an interface that enables the communication between devices and a host controller such as a computer system. It connects peripheral devices such as digital cameras, mouse, keyboards, scanners, printers, media devices, flash drives and external hard drives. It has a wide variety of uses, including support for electrical power which is why the USB has replaced a wide range of interfaces like the parallel and serial ports,. There are different types of USB ports used to connect the peripheral devices to the computer.

USB standards

- **USB 1.x:** It is an external bus standard that is capable of data transfer with the speed of 12 Megabits per second and can support up to 127 peripheral devices.
- **USB 2.0:** it is also known as **high-speed USB** a maximum transfer speed of 480 Megabits per second. However, practically it is approximately 280 Mbps.
- **USB 3.0:** it is also known as **SuperSpeed [USB](#)** and was first made available in November 2009. It is a much-improved version of USB 2.0. It supports the data transfer rate of 5 Gigabits per second, which is much faster than the speed provided by USB 2.0.
- **USB 3.1:** It is also known as **SuperSpeed+**, and it was introduced in 2013. It is almost two times faster than USB 3.0 and can transfer the data at the rate of up to 10 Gigabits per second.
- **USB 3.2:** it was introduced in 2014. It is almost two times faster than USB 3.1 and can transfer the data at the rate of up to 20 Gigabits per second.
- **USB 4:** The USB4 specification was released on 29 August 2019 by the USB Implementers Forum. The USB4 2.0 specification was released on 1 September 2022 by the USB Implementers Forum. USB4 is based on the Thunderbolt 3 protocol. It is almost two times faster than USB 3.2 and can transfer the data at the rate of up to 40 Gigabits per second.

HDMI

HDMI stands for High-Definition Multimedia Interface, and it's a standard for transmitting high-quality audio and video signals. It was developed by Hitachi, Matsushita, and Sony to replace analog audio/video cables like the DVI connector, which is still used in some applications.

HDMI STANDARDS

HDMI standards can be tabulated below with their salient features and specifications:

	Release Date	Max Bandwidth	Max Resolution	Notable Features
HDMI 1.0	2002	4.95 Gbps	1920 × 1080 @ 60 Hz	
HDMI 1.1	2004	4.95 Gbps	1920 × 1080 @ 60 Hz	DVD Audio
HDMI 1.2	2005	5.00 Gbps	1920 × 1080 @ 60 Hz	3D, HBR2, MST
HDMI 1.3	2006	10.20 Gbps	3840 × 2160 @ 30 Hz	Mini Connector, Deep Color
HDMI 1.4	2009	10.20 Gbps	3840 × 2160 @ 30 Hz	3D, Ethernet
HDMI 2.0	2013	18.00 Gbps	5120 × 2880 @ 30 Hz	HDR
HDMI 2.1	2017	48.00 Gbps	10240 × 4320 @ 30 Hz	4K at 120Hz